

TOLE BI, Kazakhstan

WMO: 382220

Lat: 43.700N

Lon: 73.783E

Elev: 456

StdP: 95.97

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-20.4	-26.3	0.4	-24.0	-22.6	0.5	-20.1	5.8	-3.3	4.6	-2.3	0.5	180	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	36.4	19.7	34.8	19.3	33.4	18.8	21.4	32.5	20.5	31.7	19.7	30.9	1.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.4	26.1	16.7	12.6	25.0	15.8	11.9	24.4	64.5	32.7	61.2	31.8	58.5	30.9	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.6	4.4	-27.2	39.0	5.0	1.2	-30.8	39.9	-33.7	40.6	-36.4	41.3	-40.0	42.2		
(5)			-27.2	23.5	4.8	1.4	-30.7	24.5	-33.5	25.3	-36.2	26.1	-39.7	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-6.1	-3.1	6.7	13.2	18.5	23.4	25.0	23.7	18.2	10.8	3.0	-4.3	(6)
(7)		DBStd	11.99	6.67	7.05	5.69	4.36	3.60	2.97	2.74	2.80	4.06	4.52	5.86	6.43	(7)
(8)		HDD10.0	1723	499	367	129	19	1	0	0	0	1	44	218	444	(8)
(9)		HDD18.3	3374	758	601	361	158	42	2	1	1	52	236	461	702	(9)
(10)		CDD10.0	2023	0	1	27	116	265	402	464	426	247	69	7	0	(10)
(11)		CDD18.3	632	0	0	0	5	47	154	206	168	48	3	0	0	(11)
(12)		CDH23.3	8529	0	0	14	132	725	1984	2722	2184	698	67	2	0	(12)
(13)		CDH26.7	4056	0	0	2	29	248	956	1434	1102	270	14	0	0	(13)
(14)	Wind	WSAvg	1.4	0.8	1.2	1.7	1.9	1.9	1.6	1.5	1.4	1.1	1.0	0.8	(14)	
(15)	Precipitation	PrecAvg	307	26	22	31	37	38	22	10	7	10	32	31	38	(15)
(16)		PrecMax	514	76	62	70	96	128	81	35	34	29	83	55	297	(16)
(17)		PrecMin	205	3	1	9	3	7	0	0	0	0	1	6	1	(17)
(18)		PrecStd	69	16	16	17	23	30	19	9	9	8	22	14	50	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	15.0	25.3	29.8	33.7	37.1	39.1	37.7	34.3	28.6	21.2	10.6	(19)
(20)			MCWB	6.0	8.6	12.8	15.8	18.0	20.3	20.5	19.9	18.7	15.1	12.1	6.4	(20)
(21)		2%	DB	6.0	10.6	21.7	26.9	31.4	35.1	36.8	35.7	31.9	25.1	16.7	7.4	(21)
(22)			MCWB	3.2	6.1	11.9	15.0	17.3	19.4	20.2	19.4	17.1	13.8	10.8	4.5	(22)
(23)		5%	DB	3.7	7.5	18.9	24.4	29.4	33.5	35.1	34.2	30.0	22.4	13.6	5.1	(23)
(24)			MCWB	1.7	4.4	10.9	14.1	16.9	18.9	19.5	18.9	16.2	12.5	8.8	2.9	(24)
(25)		10%	DB	1.9	4.9	16.0	22.1	27.5	31.9	33.4	32.5	27.6	19.8	11.0	3.3	(25)
(26)			MCWB	0.4	2.7	9.7	13.1	16.1	18.4	19.1	18.2	15.4	11.3	7.3	1.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	8.9	14.4	18.0	20.3	22.4	22.9	22.1	19.6	15.8	13.7	7.2	(27)
(28)			MCDWB	9.8	14.5	22.5	25.1	29.8	32.2	33.3	32.5	31.9	25.2	19.7	10.1	(28)
(29)		2%	WB	3.5	6.3	12.5	16.4	18.9	21.0	21.6	20.6	18.0	14.5	11.2	4.8	(29)
(30)			MCDWB	5.6	10.0	20.1	24.1	28.3	31.9	33.0	32.5	29.5	23.4	15.8	6.8	(30)
(31)		5%	WB	1.8	4.5	11.3	15.2	17.9	20.0	20.7	19.8	16.8	13.3	9.3	3.1	(31)
(32)			MCDWB	3.4	7.3	17.6	22.6	26.8	30.9	31.8	32.0	28.2	20.8	12.7	4.9	(32)
(33)		10%	WB	0.5	2.7	9.9	13.9	17.0	19.2	19.9	18.9	15.8	12.0	7.6	1.7	(33)
(34)			MCDWB	1.8	4.8	15.5	20.7	25.5	29.5	30.9	30.8	26.4	18.7	10.8	3.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	12.6	13.2	14.6	14.8	15.3	15.8	16.0	14.2	10.7	9.1	(35)
(36)			MCDBR	10.6	12.7	17.3	18.1	18.2	17.5	17.9	18.4	18.7	18.5	14.9	11.1	(36)
(37)			MCWBR	8.1	8.8	9.1	8.4	7.3	6.4	6.4	7.0	7.7	9.0	9.3	8.3	(37)
(38)		5% WB	MCDBR	9.3	12.1	15.5	15.2	15.5	15.7	15.7	16.8	17.6	15.9	12.8	10.3	(38)
(39)			MCWBR	7.2	8.6	8.8	7.7	7.1	6.5	6.4	7.0	7.9	8.3	8.4	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.357	0.397	0.443	0.466	0.450	0.446	0.437	0.405	0.386	0.416	0.399	0.352	(40)	
(41)		taud	2.085	2.025	1.978	1.967	2.073	2.129	2.168	2.239	2.239	2.062	2.046	2.088	(41)	
(42)		Ebn at Noon	725	762	779	799	826	829	832	848	833	731	663	684	(42)	
(43)		Edh at Noon	104	133	158	173	160	152	145	130	120	126	109	95	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.85	2.78	3.92	5.29	6.73	7.24	7.19	6.58	5.17	3.27	1.95	1.45	(44)	
(45)		RadStd	0.20	0.30	0.30	0.45	0.43	0.35	0.37	0.27	0.21	0.29	0.18	0.13	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-1.26	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page